Sharp subsidies for truthful production and network cooperation

Maxwell Flanagan — Claim and model comparison — 17 October 2026

Question for a specialist. Do the sharp joint coverage–incentive–budget theorems below follow from earlier work, and are the proofs and mechanism-class boundaries valid? This is an internal-review candidate, not a claim of external endorsement.

Fixed model. There are n fixed owners and d resources. Each owner reports $b_i \in \mathbb{R}^d_+$ to the same fixed ordered family $V = \{w^1, \dots, w^K\} \subseteq \mathbb{R}^d_+$:

$$f(c) = \min_{w \in V} w \cdot c, \qquad v_b(S) = f\left(\sum_{i \in S} b_i\right), \qquad T = v_b(N).$$

A single deterministic rule on every real report profile has nonnegative payments M_i, covers every coalition ($\sum_{i \in S} M_i \geq v_b(S)$), and is componentwise nondecreasing in each owner's entire bundle with others fixed. This means truthfulness for binding underreports and payment-only utility. There are no resource-use costs, overreports, coordinated-report, false-name, split/merge, other-owner or population guarantees. Owners are never replaced by strategic capacity units. A total-payment factor B corresponds to subsidy factor $B - 1$.

Proposed sharp theorem. For every integer $j \geq 2$ an eligible rule satisfies

$$\sum_i M_i \leq C_j(K) T, \qquad C_j(K) = j + \frac{\log K + 2 \log j + 1}{\log[j^3/(j^2 + j - 1)]}.$$

Choosing $j = \max\{2, \lceil \log K / [\log(2 + \log K)]^2 \rceil\}$ gives $(1 + o(1))T \log K / \log \log K$. Zero-value profiles receive zero. The worst-case optimum with *at most* K forms has the same leading constant one. The lower already holds for simple directed acyclic flow networks with at most K structural cuts and one strategic owner per edge. It is not an exact-count padding theorem for every integer K.

Upper proof to check. Delete whole owners independently with probability q; select a fixed-priority minimum *raw* form. Let p_i^q be the expected contribution capped at T conditional on deleting i, and m_i^q the expected on-minus-off production marginal. Both are own-monotone. The mixture $r_i^q = q p_i^q + (1 - q) m_i^q$ dominates a coalition-covering capped form allocation. For masked production g and total capped payment Z,

$$R = Z - g - (1 - q) g', \qquad \int_0^\eta R = \int_0^\eta Z + T - (1 - \eta) g(\eta) - 2 \int_0^\eta g.$$

A Chernoff union over the fixed forms bounds Z. The payment $M_i = j \int_0^{1/j} r_i^q \, dq$ is deterministic. Selection is uncapped; only payment contributions are capped.

Lower proof to check. Shared-fan modules keep the network simple and all auxiliary owners strategic. Actual coalition inequalities and multinomial weights $p_h \propto \lambda^h / h!$ transport only the changing owner's payment. With disjoint middle-edge and auxiliary budgets Y, Q,

$$A \leq (1 + \alpha) \mathbb{E} Y + (m/R) \mathbb{E} Q \leq \max\{1 + \alpha, m/R\} B \mathbb{E} T.$$

Taking $R = m$ preserves the constant. Cut counts and hierarchy parameters then give $\log K \sim h \log h$ and factor $(1 - o(1))h$.

New owner-count and dimension bounds. For every finite-valued monotone concave degree-one homogeneous $f : \mathbb{R}^d_+ \to \mathbb{R}_+$, including boundary-discontinuous functions, there is an eligible rule with

$$\sum_i M_i \le \min\{d,\ n\sqrt{2/(n+1)},\ 1+2\sqrt{d}\}\, T.$$

The owner bound improves the project's earlier $O(n/\log n)$ upper; the dimension bound improves d to $O(\sqrt{d})$, independently of owner count. These are upper bounds, not unrestricted sharp-rate theorems. The main paper now contains complete proofs.

Proof to check. For a fixed open-owner set B, take a supporting price on the coordinate face of $x_B = \sum_{i\in B} b_i$, extended by infinity off that face. Closed owners receive their capped support price; open owners receive $f(x_B) - f(x_B - b_i)$. This preserves coverage and own-bundle monotonicity. With missing probability q, let $C(q)$ be expected closed payments and $g(q) = \mathbb{E}f(x_B)$. Then

$$R(q) = C(q) - (1-q)g'(q), \qquad \int_0^a -(1-q)g'(q)\,dq = T - (1-a)g(a) - \int_0^a g(q)\,dq.$$

The owner bound uses $C(q) \le nqT$ and $g(q) \ge (1-q)^n T$, yielding

$$B_n(a) = na/2 + \frac{n[1-(1-a)^{n+1}]}{(n+1)a}.$$

For dimension, coordinatewise price domination and the identity

$$\mathbb{E}\sum_{i\notin B} \frac{u_i}{u_i + S_B} = \frac{q}{1-q}\Pr(S_B > 0), \qquad S_B = \sum_{i\in B} u_i,$$

give $C(q) \le 2dqT/(1-q)$, hence $D_d(a) = [1 + 2d(-a - \log(1-a))]/a$. Parameters and the choice of an entire rule depend only on fixed public n, d, K, never on the realized report profile. Exact zero conventions and relative support existence are proved in the main text.

What the new lower bounds restrict. The supplementary constructions force square-root growth for report-independent mask laws using common supporting prices selected from open reports and endpoint payments to open owners. The bounds allow asymmetric/correlated masks and randomized finite support selection. Separate results cover the maximum-coalition-marginal cap. They do not apply to arbitrary truthful payment maps or every report-dependent sampling scheme. The unrestricted lower bounds remain logarithmic over logarithmic in owner count and dimension. No matching unrestricted square-root lower bound has been established.

Priority question specific to these additions. Bachrach et al. (2009), Theorem 3.5 already establishes a square-root static cost-of-stability bound for superadditive games; the published 2018 JAIR Theorem 7 gives total factor $\sqrt{n}$. Each profile here has a nonempty core and static cost zero. Does an earlier result nevertheless imply the same bound for one own-resource-monotone rule across all fixed-owner reports? Static core existence and population monotonicity alone do not answer this question.

Closest original-source comparisons.

Production and stability. Owen (1975) and Samet–Zemel (1984) establish dual-price/core ingredients. Moretti–Norde (2021) studies the weighted multi-glove game used as a special case. Fixed-game cost of stability is established. Our issue links different reports even though each individual production game has an efficient core.

Same disclosure model. Huang–Mei–Yang–Cao (2025 v1) uses binding capacity underreports and payment utility in flow games. Its efficient mechanisms and core-selection obstruction are the nearest strategic comparison. The present theorem keeps all coalition inequalities and allows subsidy; it does not inherit their extra incentive guarantees.

Different monotonicity. Immorlica–Mahdian–Mirrokni compares an unchanged owner's payment when other owners join. Brânzei–Solymosi–Tijs (2007) compares participating subgames of unit owners (final Definition 1, Theorems 5–6). Neither condition is silently identified with our same-owner bundle comparison.

Same own-resource condition, different technology. Amigo–Belzarena–Vaton (2016 author version) defines own-resource monotonicity and a strictly concave service model. Its Theorem 1 proves Shapley own-resource monotonicity, while noting that Shapley can fail core membership. The supplement's smooth strictly concave example fits that inspected model and refutes a universal extension of the reported projection simulations, not a proved theorem. The publisher final has not been compared in full. This diagnostic is not homogeneous and is separate from the main theorem.

Established tools versus proposed contribution. Supporting prices, max-flow/min-cut, marginal contributions, own-excluded selection, concentration, weighted counting, convex support theory and uniform convergence are established. The proposed contribution is the joint sharp all-profile guarantee and matching fixed-owner simple-network lower bound. The source search is bounded; historical priority remains unconfirmed. The full technical comparison with Cao's *Superadditive Market Games* (2026, DOI 10.1002/nav.70010) remains open; only its abstract and metadata have been inspected. Owen's 1975 full original and the ABV publisher-final text also remain unreviewed. These gaps do not establish nonoverlap.

What the result does not settle. K can be exponential in an encoded technology. The remaining owner-count and resource-dimension gaps are now between logarithmic-over-logarithmic lower bounds and square-root upper bounds. Exact enumeration and the supplementary certification procedure may require exponential preprocessing. The paper asserts neither a recognized major open-problem resolution nor a general theorem for private opportunity costs.

Review evidence. The main paper contains complete complexity, owner and dimension upper proofs and the matching complexity lower. The supplement states the restricted architecture classes separately. Source-hashed internal model audits and exact rational checks are preserved in the review-update packet. These are not formal verification, specialist endorsement or priority confirmation. The 14 October PDFs and archives are historical snapshots; the 17 October package exports the main paper, supplement and this brief with their sources and scoped fresh checks. No outreach or submission has been performed.

Requested assessment. Check the cap/endpoint/deletion identities; check the reciprocal identity, boundary faces and sharp scope of mask obstructions; check symmetrization, auxiliary budgets, weighted transitions and cut inversion; identify any original theorem with the same fixed-owner, all-profile, full-coverage guarantee. The author has no affiliation. Contact details, declarations and final responsibility still need confirmation before any submission. No outreach or submission has been performed.